

ICT80004 Weekly Communication – Week #04

Student Name: Arun Ragavendhar Arunachalam Palaniyappan ID: 104837257

Organisation: Commonwealth Scientific and Industrial Research Organisation (CSIRO)

Industry Supervisor: Dr. Shigang Liu

Date Prepared: 29/08/2025 Internship Week #: 4

Day	Date	Task(s) completed
1	Monday 25 Aug 2025 8 hours	- Completed the comparative results table for vulnerable code snippets tested on GPT-4/5, Grok, Gemini, Llama-3, and DeepSeek V1. - Recorded fixes and behaviours for integer overflow, null pointer dereferences, and array index out of bounds. - Compared how different prompting styles (Chain of Thought, Action, Debate, Encouragement) affected the quality and reliability of model responses.
2	Wednesday 27 Aug 2025 8 hours	- Finalised divide by zero and SQL injection test cases with fixes, explanations, and inferences. - Consolidated results from all five vulnerability categories into the Week 3 & 4 Vulnerable Code Testing Report. - Drafted initial taxonomy of attack categories and model failure patterns, mapping tested vulnerabilities to prompt injection risks and potential defences.

Total hours completed for the week: 16

Plans for next week: #05 week (1– 5 Sept 2025)

Expand test cases to cover more vulnerabilities and refine categories.

Begin outlining the **Prompt Injection Test Suite**, mapping each case to taxonomy categories.

Add boundary and negative test cases (e.g., LLONG_MAX overflow, invalid indices, crafted SQL payloads) to confirm guard behaviour.

Acknowledgment / Appreciation mail from supervisor for work done for week #01 – week #03

Weekly Communication / Reflection update - #03 week, 18– 22 Aug 2025

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Arun Ragavendhar

Dear Dr. Shigang Liu, I hope you are well. Please find below my update for Week 3 of the internship. Activities completed this week (Total: 16 hours): - Continu

Sun 24 Aug, 16:22 (7 days ago)

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L

Liu, Shigang (Data61, Clayton)

to me

Hi Arun,

Thanks for your effort. It looks good to me.

Kind regards,

Shigang

Tue 26 Aug, 09:46 (5 days ago)

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Screenshot of Timely EMAIL communication update to the Supervisor at the end of week #04 sent on 29 August 2025 Friday at 3:00 PM (15:00).

Weekly Communication / Reflection update - #04 week, 25 – 29 Aug 2025

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Arun Ragavendhar <arunragavendhar.1999@gmail.com>

to Shigang

29 Aug 2025

Dear Dr. S

I hope you

Activities

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from: Arun Ragavendhar <arunragavendhar.1999@gmail.com>

to: "Liu, Shigang (Data61, Clayton)" <Shigang.Liu@data61.csiro.au>

date: 29 Aug 2025, 15:00

subject: Weekly Communication / Reflection update - #04 week, 25 – 29 Aug 2025

mailed-by: gmail.com

ok, Gemini, Llama-3, i

Documented secure fixes and model behaviours for five vulnerability categories: integer overflow, null pointer dereference

Weekly Communication / Reflection update - #04 week, 25 – 29 Aug 2025

A

Arun Ragavendhar <arunragavendhar.1999@gmail.com>

to Shigang

Fri 29 Aug, 15:00 (2 days ago)

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Dear Dr. Shigang Liu,

I hope you are well. Please find below my update for Week 4 of the internship.

Activities completed this week (Total: 16 hours):

- Completed the comparative results table for vulnerable code snippets tested across GPT-4/5, Grok, Gemini, Llama-3, and DeepSeek V1, using four prompting styles (Chain of Thought, Chain of Action, Chain of Debate, and Encouragement Learning).
- Documented secure fixes and model behaviours for five vulnerability categories: integer overflow, null pointer dereference, array index out of bounds, divide by zero, and SQL injection.
- Analysed how each model and prompt style influenced the fix. For example, GPT-4/5 and Llama-3 consistently produced minimal safe guards, while Gemini occasionally suggested unsafe alternatives such as type changes or clamping, which were rejected.
- Consolidated all findings into the **Week 3 and Week 4 Vulnerable Code Testing Report**, with detailed tables, explanations, and inferences.
- Started drafting the taxonomy of attack categories and model failure patterns, linking the vulnerabilities we tested to broader prompt injection risks and possible defence methods.

Plan for next week (1–5 Sept 2025):

- Extend the test cases to cover additional vulnerabilities and generalise across more categories.
- Outline the initial structure of the **Prompt Injection Test Suite**, mapping each test case to the taxonomy.
- Add boundary and negative tests (e.g., LLONG_MAX, invalid array indices, and crafted SQL payloads) to validate whether models reliably enforce safety.
- Prepare a summary presentation of the Week 3–4 results and taxonomy progress for review and discussion in the upcoming meeting.

Kind regards,
Arun Ragavendhar Arunachalam Palaniyappan
ICT80004 Internship Student – CSIRO Data61

2 attachments • Scanned by Gmail ⓘ

